

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

ANSWER SCHEME

CLASS

INDEX NUMBER

BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

11 September 2017

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	25
2	25
Section B	
3 OR 4	25
Total	75

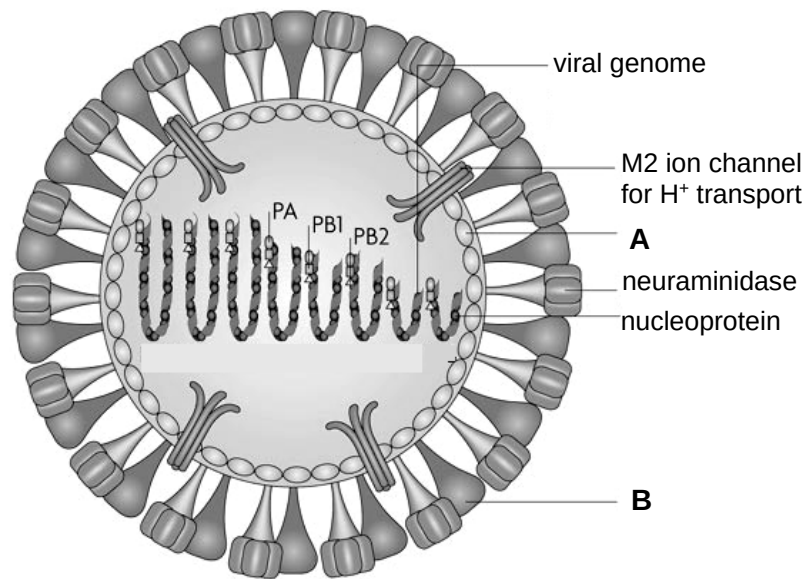
This document consists of **18** printed pages.



Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows the influenza virus.

**Legend**

PA, PB1, PB2 are RNA polymerases

Fig. 1.1

- (a) (i) Identify structures **A** and **B**.

A *matrix prot*

B *haemagglutinin*

[2]

- (ii) Explain why the influenza virus needs its own RNA polymerase.

1. needed to transcribe negative sense viral RNA into positive sense viral RNA

to act as mRNA for translation into new viral prot / templates for syn of new viral genome

2. host cell RNA pol transcribes DNA template to RNA / is DNA-dep RNA pol

virus need RNA-dep RNA pol

[2]

- (b) With reference to B and T cells, describe the adaptive immune response upon primary exposure to influenza virus.

1. influenza virus is engulfed by macrophages / dendritic cells

Ag is processed into short peptides / epitope

2. Ag / epitope is presented on MHC-II to Th / CD4+ cell at TCR

Th cell is activated

3. activated Th cell secretes interleukin / cytokines

and bind B cells (with Ag at MHC-II) to activate it

4. activated B cells proliferate and differentiate into plasma cells

that secretes Ab that binds Ag to precipitate / opsonise / neutralise virus

5. activated Tc cells enz / chemicals / perforin

to trigger apoptosis / cause lysis of / kill infected cells

[5]

(c) Vaccine for influenza is readily available in many countries including Singapore. It is advised that members of the public get vaccinated yearly.

(i) Describe the benefits of vaccination.

1. confer immunity to individuals not previously exposed to the virus

prevents / protects against death / illness / disabilities caused by disease

2. confer immunological memory / long term / lifetime immunity to individual

due to production of memory B and T cells

3. reduce spread of virus within human popn

by conferring herd immunity / since virus relies only on human as host

[3]

(ii) Describe what happens when a person vaccinated for influenza is subsequently infected with the same strain.

1. more rapid / lag phase is shorter, more intense & prolonged immune response / high & steady levels of Ab

due to immunological memory

2. memory B cells rapidly undergo clonal expansion → plasma cells → large no. of Ab which bind and inactivate virus

Ab produced remain in circulation longer to ensure infection is eliminated OR

BCR has higher affinity for Ag thus responding more rapidly

[2]

(iii) Explain why yearly vaccination is recommended for influenza.

1. due to antigenic drift → accumulation of mutations in the viral genome during replication ® antigenic shift (does not occur as frequently)

due to the lack of proofreading mechanisms of RNA pol

2. leading to Δ in 3D config of HA & NA

which is no longer recog / complementary to Ab, BCR & TCR on memory cells

[2]

Doctors sometimes prescribe antibiotics to patients who are infected by influenza to combat secondary bacterial infections. Gramicidin A is an example of an antibiotic. Fig. 1.2 shows the molecular structure of Gramicidin A.

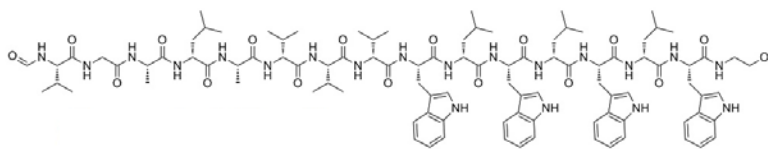


Fig. 1.2

(d) Illustrate the reaction that forms the peptide bonds between two amino acids.

1. **indicate carboxylic grp + amine grp as rxting grps**
 2. **condensation**
 3. **loss of H_2O**
 4. **label peptide bond in dipeptide ($C=O$ and NH must be in trans position)**
- (any 2 MP for 1m)

[2]

Gramicidin A folds into a 3-dimensional configuration that inserts itself into the bacterium's cell surface membrane. It allows non-specific movement of ions which eventually cause the bacterial cell to die. Fig. 1.3 shows the interaction of Gramicidin A with the bacterium's cell surface membrane.

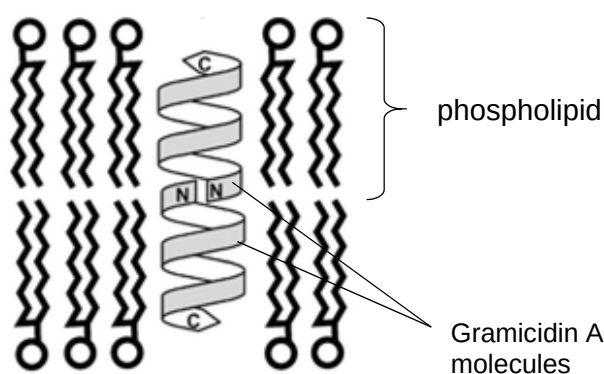


Fig. 1.3

(e) (i) Describe how the Gramicidin A shown in Fig. 1.2 folds into the 3-dimensional structure shown in Fig. 1.3.

1. **forms α -helix of spiral shape (with 3.6 aa per turn)**

held by H bonds

2. **btw H of NH and O of CO of peptide bonds**

$n+4$ aas away

3. **2 molecules assoc. with each other at N-ter**

folded such that hydrophobic R grps face o/s to interxt with FA tails / hydrophilic R grps in/s facing channel to interxt with ions

[2]

- (ii) Using the information provided and Fig. 1.3, explain how Gramicidin A kills the bacterium.

1. Gramicidin A forms hydrophilic channel

allows non-specific / unregulated movement of ions in or out of bacterium (down conc. grad.)

2. disrupting ionic balance in the bact cell

cause disruption metabolic fn leading to cell death

[2]

® osmotic lysis as cell walls are not weakened

- (f) Upon prescription of antibiotics, doctors often advise patients to complete the course of antibiotics even if symptoms of disease have ceased. One of the reasons cited was that not completing the course of antibiotics may increase the chance of antibiotic resistant bacteria to evolve.

With reference to natural selection, explain the basis for the need to complete the prescribed course of antibiotic.

1. not completing prescribed course of antibiotics may leave small nos. of bact remaining

a mutation that confers antibiotic resistance may occur in the remaining bact popn

2. presence of antibiotics (in patient's circulation) act as a selection pressure

selecting for mutant resistant strain with favourable phenotype which experience 1er survival rate & repro success

3. thus completing the antibiotic course ensures that all susceptible bact in the patient dies

and leave no bact popn for mutation to occur in presence of antibiotic in the bact's env

4. mutant resistant bact strain may not be selected for in env without antibiotics

as resistant strains may be outcompeted by susceptible strains due to o/r selection pressures

[3]

(any 3 MP)

[Total: 25]

- 2 A researcher investigated the pathway by which carbon dioxide is converted to organic compounds during photosynthesis. The apparatus used is shown in Fig. 2.1.

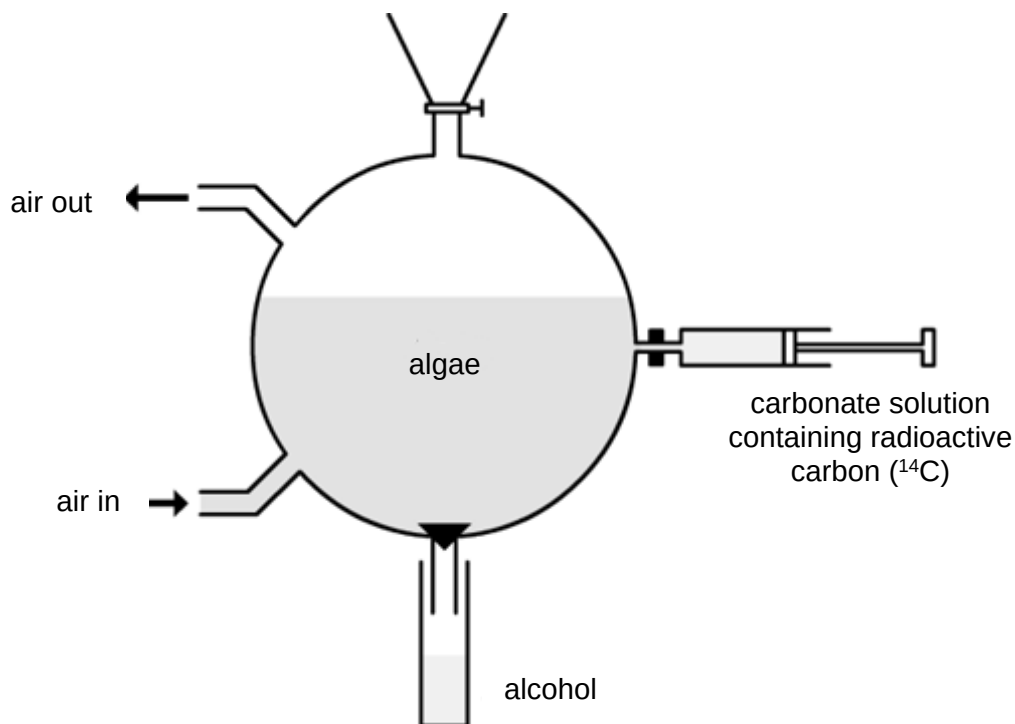


Fig. 2.1

While the apparatus was in the dark room, the researcher supplied the algal cells with $^{14}\text{CO}_2$. The contents of the apparatus were thoroughly mixed and light was switched on subsequently. At five-second intervals, a few of the cells were released into hot alcohol, which killed the cells very quickly. The intermediates of the reactions were subsequently analysed and the chromatograms in Fig. 2.2 showed the results of the analysed intermediates by chromatography (a separation technique) at different timings.

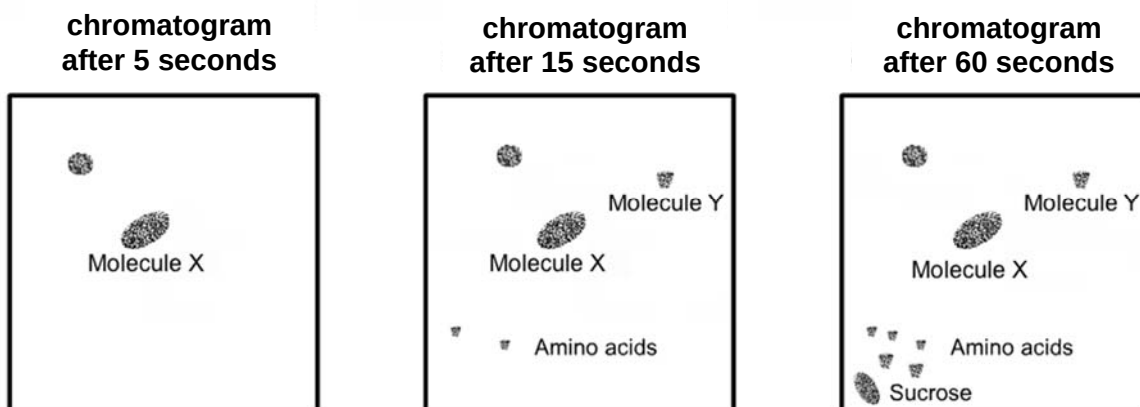


Fig. 2.2

- (a) (i) Identify molecules X and Y.

glycerate-3-phosphate;;

X ***@glycerate phosphate***

@3-phosphoglycerate

® short form

Y ***glyceraldehyde-3-phosphate;;***

@triose phosphate

[2]

(ii) Explain your answer in part (i).

1. $^{14}\text{CO}_2$ combines with 5C RuBP during carbon fixation

catalyzed by Rubisco / RuBP carboxylase;;

2. forming an unstable 6C intermediate

which is broken down to form molecule X, 3PGA (with ^{14}C incorporated);;

3. 3PGA is phosphorylated to form 1,3-bisphosphoglycerate

and undergoes reduction;;

4. using red NADP

to form molecule Y, G3P/ TP (with ^{14}C incorporated);;

[4]

Part (i) must be correct in order to be awarded full marks for part (ii)

(iii) Explain why sucrose and amino acids are identified in the chromatogram only after 60 seconds.

1. with every 6 G3P/ TP (with ^{14}C incorporated) produced, 1 G3P/ TP exits the Calvin cycle

to be converted into other carbohydrates and organic compounds (with ^{14}C incorporated);;

[1]

Dinitrophenol is a metabolic poison that can embed within the thylakoid membranes of chloroplasts and provide an alternate route for H^+ to diffuse across the thylakoid membranes.

(b) Explain how the concentration of intermediates of the Calvin cycle is affected by dinitrophenol.

**1. less H^+ diffuse through ATP synthase
@proton gradient less steep**

less ATP produced (during light-dept rxn);;

2. less ATP for use in Calvin cycle

during carbon reduction and regeneration of RuBP;;

3. accumulation of 3PGA

↓ production of 1,3-bisphosphoglycerate / G3P/ TP/ RuBP;;

[3]

Another experiment was carried out by another student to determine the concentration of carbon dioxide in the leaves of plants at different times of the day. The results are shown in Table 2.3

Table 2.3

Mean carbon dioxide concentration (ppm)	
8pm to 4am	8am to 4pm
328	106

- (c) Using knowledge on Calvin cycle and Krebs cycle, account for the difference in concentration of carbon dioxide in the leaves for the two periods shown in Table 2.3.

1. more CO₂ produced from 8pm to 4am compared to 8am to 4pm

QV;;

2. 8pm to 4am – no light

light-dept rxn of photosynthesis ⊗ occur ➔ ⊗ produce ATP and red NADP;;

3. Calvin cycle ⊗ occur ➔ CO₂ ⊗ used up during carbon fixation

CO₂ produced in Krebs cycle during oxidative decarboxylation;;

4. 8am to 4pm – presence of light

CO₂ produced during Krebs cycle taken up during Calvin cycle;;

[4]

(accept reverse argument)

- (d) Explain the effect of higher concentration of carbon dioxide on the rate of carbon fixation during the period 8am to 4pm.

1. ↑ CO₂ ➔ ↑ [S]

↑ freq of effective collisions between CO₂ (& RuBP) and active site of Rubisco;;

2. ↑ ES complex formation per unit time ➔ ↑ products formed (i.e. 3PGA) per unit time

↑ rate of carbon fixation;;

[2]

Studies were carried out on soil-dwelling aerobic and anaerobic bacteria. Samples were taken from different depths at intervals of one month and six months after the soil was put into a large heap for storage.

Table 2.4 shows the numbers of aerobic and anaerobic bacteria at different depths in the stored soil.

Table 2.4

depth in soil store / m	mean number of bacteria per gram of stored soil $\times 10^7$			
	aerobic bacteria		anaerobic bacteria	
	after one month	after six months	after one month	after six months
0.0	12.4	12.5	0.4	0.6
0.5	10.1	8.3	0.6	1.0
1.0	9.8	5.9	0.8	3.8
1.5	9.7	3.1	0.8	7.6
2.0	10.5	0.8	0.7	8.1
2.5	10.8	0.7	0.8	8.5
3.0	10.2	0.9	0.6	8.8

- (e) (i) Account for the trends shown by the distribution of the two types of bacteria after six months.

1. aerobic bacteria decrease with depth from mean number of bacteria per gram of stored soil of 12.5×10^7 to 0.9×10^7

anaerobic bacteria increase with depth from mean number of bacteria per gram of stored soil of 0.6×10^7 to 8.8×10^7 ;;

2. aerobic bacteria requires oxygen for respiration

oxygen content of soil decreases with depth;;

3. less oxygen available as the final electron acceptor in ETC

decrease in ATP synthesis for use in cellular functions;;

[3]

- (ii) Suggest how aerobic bacteria are structurally adapted for cellular respiration.

1. presence of cytoplasmic membranes in aerobic bacteria

increase surface area;;

2. for more embedding of electron transport chain and ATP synthase

to allow electron transfer / to drive oxidative phosphorylation;;

[2]

In a further study, soil samples were taken at two depths, **A** and **B**, in the soil store. The samples were taken at intervals over six years. Soil samples of equal mass were used to determine the activity of dehydrogenases in aerobic bacteria.

Fig. 2.5 shows the mean dehydrogenase activity of the bacteria in these samples.

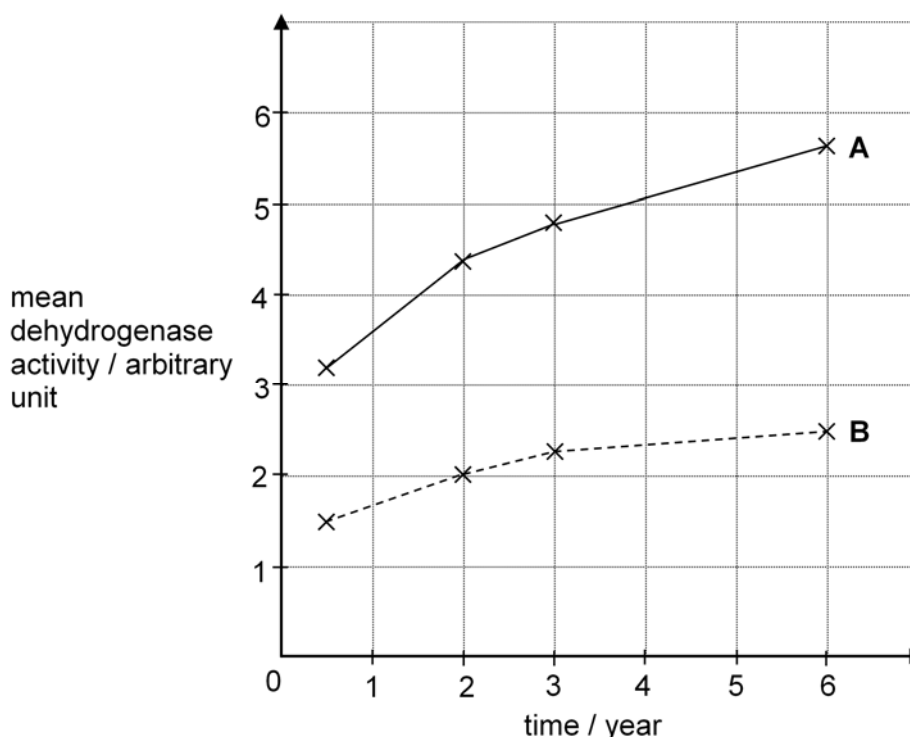


Fig. 2.5

- (f) (i) State with evidence from Fig. 2.5 which depth, **A** or **B**, were samples taken from a greater depth.

1. **depth B;;**

2. **bacteria in samples taken from depth B shows a lowered mean dehydrogenase activity with 1.5 to 2.4 au**

compared to 3.2 to 5.7 au in samples taken from depth A;;

[2]

- (ii) Explain the roles of dehydrogenase in Krebs cycle of the aerobic bacteria.

1. **catalyze oxidation reactions in Krebs cycle;**

loss of protons and electrons;

2. **reduction of NAD and FAD;**

to form reduced NAD and reduced FAD;

[2]

[Total: 25]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts (a), (b), etc., as indicated in the question.

- 3 (a)** Greenhouse gases are key contributors to climate change affecting animals and plants in the environment they live in.

Discuss the effects of climate change in the global environment. [13]

- 1. due to increase in global ave temp as GHGs trap heat from sun's radiation causing accumulation of heat in atm;;**
- 2. results in melting of polar ice caps causing faster water surface run-off on land towards the sea (reduction of water infiltrating into aquifers as freshwater supplies & contribute to rise in sea levels);;**
- 3. results in rising sea levels due to warming of ocean waters leading to vol. expansion;;**
- 4. due to melting of ice (ice sheets & sea ice) resulting in loss of habitat & food source for animals such as polar bears;;**
- 5. results in reduction of freshwater supplies due to saltwater intrusion where salt water mixes with freshwater stores;;**
- 6. due to increased demand from humans as popn increases & increased freq & duration of droughts;;**
- 7. results in more freq. extreme unpredictable weather events such as heat waves & heavy rains/ higher evaporation rate in water cycle resulting in faster cloud formation (major fluctuations in precipitation – rain, snow);;**
- 8. causing flash floods on land surface leading to large vol of water surface run-off towards large bodies of water/ sea causing effects to residential areas, loss of crops, cattle;;**
- 9. results in death of coral reefs due to coral bleaching (corals lose their colours back to their white/brown original state);;**
- 10. loss of symbiotic algae (zooxanthallae) from coral tissue & dissolving of coral skeleton;;**
- 11. due to ocean acidification by absorption of carbon dioxide by ocean waters to form carbonic acid;;**
- 12. results in habitat migration of fish & insects to higher latitudes & altitudes with cooler temp to survive & thrive;;**
- 13. loss of biodiversity in current location, affecting food web (loss of prey for predators), change in plant distribution & adaptations;;**
- 14. increase in competition for resources with natives, increase in species variety in new habitat;;**
- 15. results in release of carbon dioxide & methane;;**

16. *from melting frozen organic matter (permafrost) in soil due to decomposition by microbes;;*
17. *increase in metabolic rate among insects such as mosquitoes due to higher kinetic energy b/w enzymes & substrates;;*
18. *shorter life cycle, faster devt (mature faster) & reproduction of mosquitoes, higher population, more transmission of dengue virus as Aedes aegypti are vectors, spread of dengue;;*
19. *spread to temperate regions (now exposed) not only tropical regions due to migration of mosquitoes;;*

- (b) After viruses infect host organisms, they are able to make use of host cell machinery to replicate and reproduce thereby causing diseases in the host organism.

Describe how dengue causes viral disease in humans.

[12]

Infection of host cell (reproductive cycle of DENV)

1. *infected mosquito injects DENV into bloodstream infecting keratinocytes & dendritic cells;;*
2. *E glycoprot of DENV binds to receptors on host cell & enters host cell via receptor-mediated endocytosis (RME);;*
3. *acidification of endosome → conformational Δ where fusion of viral env with endosomal memb → release of nucleocapsid into host cytoplasm;;*
4. *viral RNA translated by host ribosomes on rough ER → produce viral polypeptides which are cleaved by host & cellular protease to pdc 10 prots;;*
5. *viral RNA is transcribed to form –ve sense RNA which acts as templates for synthesis of more viral genome;;*
6. *viral RNA associate with capsid prots forming nucleocapsid at surface of rER, nucleocapsid buds into ER forming an env containing E & M glycoprots on surface;;*
7. *immature viruses travel through Golgi body & undergoes maturation where furin cleaves b/w pr & M prots;;*
8. *fusion of vesicle memb with host cell memb releasing mature virions to infect other cells;;*

Pathogenesis (devt of disease)

9. *infected dendritic cells travel to lymph node & present viral Ag → activate monocytes → monocytes infected;;*
10. *monocytes travel to site of infection via lymphatic system infecting more cells → viremia (presence of viruses in blood) causing fever;;*
11. *infection & apoptosis of monocytes & macrophages causes low WBC count/ leukopenia;;*
12. *infection & apoptosis of endothelial cells resulting in thinning/ weakening of endothelium lining blood vessels → haemorrhage (escape of blood from ruptured blood vessel);;*

[Total: 25]

- 4 (a) Effector molecules are responsible for the regulation of transcriptional units in prokaryotes.

Using named examples, explain the roles of these effector molecules in the negative feedback regulation of transcriptional unit in a prokaryote such as *Escherichia coli*.

[12]

lac operon: allolactose

1. **effector molecule: allolactose (an inducer) for lac operon, formed from isomerization of lactose by β -galactosidase;;**
2. **lactose from env bacteria tpted into bacteria via lactose permease;;**
3. **allolactose binds to active lac repressor prot bound to operator of lac operon $\rightarrow$ changes 3D config of lac repressor, no more complementary to operator site;;**
4. **lac repressor dissociates from operator site, RNA pol binds to promoter site;;**
5. **initiates transcription of structural genes, lacZ, lacY & lacA, lac operon is switched on;;**
6. **in absence of allolactose, lac repressor remains bound to operator site, RNA pol cannot access promoter site/ blocked from initiating transcription, lac operon is switched off;;**

trp operon: tryptophan

7. **effector molecule: tryptophan (a co-repressor) for trp operon, synthesised when trp operon is switched on;;**
8. **tryptophan present in bacteria env will bind to trp repressor, $\rightarrow$ changes 3D config of trp repressor becoming active;;**
9. **trp repressor DNA binding site complementary to operator site, binds to operator site preventing RNA pol binding;;**
10. **no initiation of transcription of structural genes trpE, trpD, trpC, trpB & trpA, trp operon is switched off;;**
11. **in absence of tryptophan, trp repressor is inactive, trp repressor does not bind to operator site, RNA pol can bind to promoter site,**
12. **initiates transcription of structural genes to form mRNA to synthesise enzymes for biosynthesis of tryptophan, trp operon is switched on;;**

- (b) Protein production in eukaryotes is controlled at all stages of the process.

Explain how protein production is controlled in eukaryotes and the advantages of regulating protein production at different stages. [13]

Stages of Protein Production Control (max 9m)

chromatin remodelling/ chromatin level regulation (max 2m)

1. **DNA methylation** → covalent addition of a methyl grp to DNA catalysed by DNA methyltransferase decreases transcription rates;;
2. **histone acetylation** → adding an acetyl grp to lysine residues at N-terminal of histone tails, catalyzed by HATs increases transcription rate;;

OR **histone deacetylation** of histone involves removal of an acetyl grp from lysine residues at N-terminal of histone tails catalysed by HDACs decreases transcription rate;;

transcription control (max 2m)

3. **core promoters** (TATA box) → found upstream of gene bound by general TFs (trans-acting element) & RNA pol forming transcription initiation complex to initiate transcription;;

proximal promoters (CAAT/ GC box) → found further upstream of core promoters bound by various TFs to promote transcription;;

4. **enhancers** (distal acting element) → regulatory DNA seq bound by activators (specific TF) → enhance transcription initiation;;

silencers (distal acting element) → regulatory DNA seq bound by repressors (specific TF) → inhibit activators & reduce transcription initiation;;

5. **trans-acting elements** located on diff chromosome as gene they regulate → code for specific TFs (activators & repressors);;

post-transcription control (max 2m)

6. **5' capping** → methylated guanine added to 5'end pre-mRNA forming a 5' cap;;

3' polyadenylation → multiple adenine residues added to 3'end of pre-mRNA forming a poly(A) tail at AAUAAA seq (polyadenylation signal);;

7. **RNA splicing (constitutive & alternative)** → introns removed & exons joined by spliceosomes at splice sites;;

translational control (max 2m)

8. **mRNA stability** → mRNA stability depends on length of poly(A) tail → 5' cap of mature mRNA poly(A) tail shortened by exonucleases in cytoplasm resulting in mRNA degradation, length of poly(A) tail can be lengthened by cytoplasmic enz to increase its lifespan;;

9. **initiation of translation** → blocked by regulatory prots that bind to 5' or 3' UTR) prevents attachment of small subunit of ribosomes to initiate translation initiation complex;;

10. **miRNA** regulates gene expression in cytoplasm by repressing translation of mRNAs and/or degrading mRNAs → miRNA complexes with RISC → binds

to target mRNA & degrades it/ binds to 3' UTR leading to an inhibition of translation;;

post translational control (max 1m)

11. **protein stability** → prots undergoing proteolytic degradation conjugated with ubiquitin → recognised & degraded by proteasome → lower conc of prot needed;;
12. **protein processing** → prots become functional via cleaving of prots (e.g. cleaving of preproinsulin to form insulin, biochemical modification (e.g. phosphorylation/ glycosylation of prots));

Advantages of Regulating Protein Production (max 4m)

chromatin remodelling/ chromatin level regulation

13. switching genes on & off to restrict active genes to those only req'd by specific cells, more efficient/ less wasteful of resources;;

transcriptional level

14. allows regulation of rate of prot pdtn, match short term requirements/allow flexibility;;

post-transcriptional level (max 1m)

15. allows for pdtn of diff prot variants from same pre-mRNA via alternative splicing/ increases coding capacity of genome;;
16. facilitate export of mRNA to cytoplasm → prevent enzymatic degradation of mRNAs by exonucleases → controls mRNA stability;;
17. removal of introns prevents extra aas from being incorporated during translation;;

translational level

18. affecting timeliness of prots to be translated from mature mRNA depending on conc of prots present in cell;;

post-translational level

19. allows rapid pdtn of active prot from inactive form via phosphorylation & glycosylation where it is needed, safe transport/ storage of produced prots when not needed immediately;;

QWC

scientific argumentation exemplified by two or more advantages of regulating protein production linked coherently to the correct stage of the process

[Total: 25]

